

RESEARCH ARTICLE

Associations between psychological resilience and metrics of white matter microstructure in pediatric concussion

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Abstract

This study investigated associations between psychological resilience and characteristics of white matter microstructure in pediatric concussion. This is a case control study and a planned substudy of a larger randomized controlled trial. Children with an acute concussion or orthopedic injury were recruited from the emergency department. Participants completed both the Connor–Davidson Resilience Scale 10 and an MRI at 72 h and 4-weeks post-injury. The association between resiliency and fractional anisotropy (FA), mean diffusivity (MD), axial diffusivity (AD), and radial diffusivity (RD) at both timepoints were examined. We examined whether these associations were moderated by group. The association between resiliency captured at 72 h and diffusion tensor imaging metrics at 4 weeks was also investigated. Clusters were extracted using a significance threshold of threshold-free cluster enhancement corrected $p < .05$. A total of 66 children with concussion (median (IQR) age = 12.88 (IQR: 11.80–14.36); 47% female) and 29 children with orthopedic-injury (median (IQR) age = 12.49 (IQR: 11.18–14.01); 41% female) were included. A negative correlation was identified in the concussion group between 72 h resilience and 72 h FA. Meanwhile, positive correlations were identified in the concussion group with concussion between 72 h resilience and both 72 h MD and 72 h RD. These findings suggest that 72 h resilience is associated with white matter microstructure of the forceps minor, superior longitudinal fasciculus, and anterior thalamic radiation at 72 h post-concussion. Resilience seems to be associated with neural integrity only in the acute phase of concussion and thus may be considered when researching concussion recovery.

KEYWORDS

concussion, diffusion tensor imaging, pediatric, resilience, white matter microstructure

1 | INTRODUCTION

Children and adolescents are among those with the highest rates of traumatic brain injury related emergency department visits, with

increased vulnerability to neuronal damage (Keightley et al., 2012; McCrory et al., 2004; Taylor et al., 2017). Although concussions are considered as mild traumatic brain injuries, those with a concussion can experience a variety of somatic, emotional, and cognitive

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symptoms, that can last for months to years, impacting quality of life. (Beauchamp et al., 2019; McCrory et al., 2017; Novak et al., 2016; Yeates, 2010). Concussion is characterized by neural deficits such as decreases in white matter structural integrity, altered patterns of brain activity, dysfunctional autoregulation and impaired vascular reactivity, as well as depressed cerebral blood flow (Bigler & Bazarian, 2010; Choe et al., 2012; Johnson et al., 2012; Maugans et al., 2012; Mayer et al., 2011; Roberts et al., 2014; Werner & Engelhard, 2007).

Psychological resilience is defined as an individual's ability to adapt or thrive in the face of trauma or adversity that may threaten healthy functioning or development (Masten & Barnes, 2018). The construct of resilience is an abstraction of all adaptive capacities available to an individual to successfully cope with various stressors or traumas (Masten & Barnes, 2018). High levels of negative stress can impact children's emotions, behavior, physical health, and socialization (Barnes, 2016). Highly resilient children are more likely to successfully cope with stress, decreasing its effects on their functioning (Barnes, 2016). Thus, resilience may be considered as a measure of the ability to cope with stress (Connor & Davidson, 2003). Resilience has also been characterized as a process, and not a trait. Measures such as the Connor-Davidson Resilience Scale (CD-RISC), were designed with the intention to assess the modifiability of resilience. Previous studies using this assessment demonstrated that resiliency scores can be modifiable and can improve with appropriate treatments (Connor & Davidson, 2003; Davidson et al., 2005). Hence, children that follow such treatments may experience the benefits of high resilience on well-being. If highly resilient children have differential white matter integrity following concussion, treatments targeted toward enhancing resilience may bolster brain structure post-concussion. This provides an opportunity for further studies to investigate whether specific treatments enhance resilience in pediatric concussion, and to examine such effects on brain health.

There is emerging evidence that would suggest psychological resilience is associated with favorable behavioral and, possibly, neural outcomes in clinical pediatric cases, particularly traumatic brain injury. Two recent studies of children with acute concussion, have demonstrated that low resilience was associated with high post-concussion symptom severity and was often accompanied by both high depressive and anxiety symptoms (Bunt et al., 2021; Durish et al., 2018). The association between resilience and concussion symptomatology was also determined to be mediated by depressive and anxiety symptoms (Durish et al., 2019). With regards to neural outcomes, a recent report indicated that fractional anisotropy (FA) and mean diffusivity (MD), measures of white matter integrity, were associated with resilience-promoting factors (i.e., intact intellectual functioning, access to community resources, presence of a warm and supportive caregiver, and prosocial peer influences) in complicated mild, moderate and severe TBI (Schmidt et al., 2021). Currently, no studies have examined the white matter microstructural correlates of resilience in pediatric concussion, or mild traumatic brain injury, in the acute and subacute phase (<1-month post-injury). The subacute phase of concussion is often associated with significant changes in FA, MD, and other measures of white matter integrity (Borich et al., 2013; Mayer et al., 2012;

Roberts et al., 2014; Wu et al., 2010). The neural correlates of resilience may be a protective factor in this critical phase of the pediatric concussion recovery process. If highly resilient children experience diminished neural deficits following concussion, it may be postulated that resilience-targeted treatments may alleviate such deficits in children with lower resilience and improve brain health. Investigating both behavioral and neural correlates of resilient children might help further our understanding of recovery processes in the subacute phase of pediatric concussion.

Diffusion tensor imaging (DTI) may serve as an effective technique to identify white matter correlates of resilience in children with concussion (Bigler & Bazarian, 2010; Gardner et al., 2012; Roberts et al., 2014). Alterations in microstructural white matter integrity following pediatric concussion can be detected immediately following the injury (acute phase; Roberts et al., 2014) and can persist beyond 3 months post-injury (chronic phase; Roberts et al., 2014; Ware, Yeates, Geeraert, et al., 2022; Ware, Yeates, Tang, et al., 2022). The most common measure used to assess white matter integrity with DTI is FA, a normalized measure of the directionality of molecular diffusion. An increase in myelination or axonal density decreases intercellular water diffusion, thus increasing FA (see Feldman et al., 2010 for a review). Additional information for the characterization of white matter microstructure can be extracted from DTI measures, including axial diffusivity (AD), that is diffusivity parallel to the axis of diffusion, radial diffusivity (RD), that is diffusivity perpendicular to the axis of diffusion, and MD, the magnitude of diffusion. Traumatic brain injury animal model research provides further insight to DTI metric correlates of white matter microstructure. In mouse models of white matter injuries, increased FA with decreased AD and RD is thought to reflect axonal degeneration, while decreased FA with increased RD alongside unchanged AD may reflect demyelination (Budde et al., 2008, 2009).

Healthy children and adolescents experience ongoing microstructural development of fiber tracts, including myelination progression (Asato et al., 2010). Such microstructural development of white matter can be detected using DTI metrics. Several studies reported a gradual decrease in MD alongside a gradual increase in FA across numerous major fiber tracts in developing brains (see Qiu et al., 2015 for a review). Maximal levels of FA along various fiber tracts are achieved at different ages (Lebel et al., 2008). Hence, differences in DTI metrics are detectable in pediatric samples that are still undergoing white matter development. Children and adolescents are thought to have increased risk of neuronal injury from concussion, partly due to ongoing myelination (Keightley et al., 2012; Kirkwood et al., 2006; McCrory et al., 2004). Fiber tracts that are not fully developed, or with lower extent of protective myelin, likely have diminished shear resistance (Thibault & Margulies, 1998). Although the developing brain was thought to be more adaptive due to its plasticity and myelination, some evidence suggests that deficits in white matter microstructure can persist for years following concussion (Caeyenberghs et al., 2011; Roberts et al., 2014).

In pediatric and young adult concussion samples, however, some research has detected an increase in FA in the acute phase following concussion, which might reflect axonal swelling or cytotoxic edema

(Bigler & Bazarian, 2010; Gardner et al., 2012; Roberts et al., 2014; Wu et al., 2010). This variability in the interpretation of what FA values indicate suggests that FA alone is not sufficient to characterize microstructural white matter integrity, at least within the context of concussion. A meta-analysis conducted by Roberts et al. (2014) reports a trend of increases in FA and decreases in MD within the acute and subacute phases of pediatric concussion. In chronic pediatric concussion, however, global decreases in FA and increases in MD can be identified, suggesting depressed integrity of white matter. Such changes in DTI measures were also associated with increases in pediatric concussion-related symptoms (Bartnik-Olson et al., 2014). Further research is required to accurately assess the implications of the direction of changes in DTI metric values, particularly in pediatric concussion, while also considering whether resilience is associated with the outcomes as such.

Exploring the neural correlates of resilience in pediatric concussion is a necessary precursor to understanding the capacity of resilience-targeted treatments for improving brain health following concussion. It is paramount for studies to first explain the effects of resilience on brain structure and function after a concussion, as research has already demonstrated positive effects of resilience on clinical outcomes (Durish et al., 2018; Durish et al., 2019). If high resilience is associated with different neural outcomes in the event of a concussion, resilience-targeted treatments may further benefit current guidelines for optimal concussion recovery.

The objectives of this article were to investigate self-reported resiliency as a correlate of FA, MD, AD, and RD maps assessed at both 72-h and 4-weeks post-injury in pediatric and orthopedic-injured children. We also sought to determine whether these correlates were moderated by type of pediatric injury (concussion or orthopedic). A secondary objective was to assess 72 h resiliency as a correlate of 4-week FA, MD, AD, and RD maps in the concussion group.

Associations between resilience promoting factors and DTI metrics were previously identified in youth with concussion (Schmidt et al., 2021). Similarities in DTI metric alterations following injury were reported between children with concussion and orthopedic-injured children, likely attributable to premorbid or injury-risk-related factors (Hulkower et al., 2013). Consequently, we hypothesized that resilience scores are associated with DTI metrics in both groups at both timepoints. Specifically, we anticipated that resilience was positively correlated with FA and AD values, while negatively correlated with MD and RD values in both groups at each timepoint. We also hypothesized that associations of resilience and DTI metrics at both timepoints would not differ between groups. In other words, there are no interactions between group and resilience when predicting DTI metrics.

2 | METHODS

2.1 | Design

This was a substudy of the randomized clinical trial entitled Pediatric Concussion Assessment of Rest and Exertion (PedCARE; Ledoux

et al., 2019), where a subsample of children and adolescents with a concussion received an MRI at <72 h (± 48 h) and at 4 weeks (± 5 days) from enrolment in the emergency department, post-injury. PedCARE studied two groups of participants with a concussion based on rest until asymptomatic or exertion after 72 h of rest instruction but found no differences in the overall group means of physical activity (Ledoux et al., 2021). For this reason, the current study combined both concussion groups together. Participants with an orthopedic injury were also included as a comparison group, allowing for comparisons both within as well as between groups. Participants were included in this substudy only after satisfying inclusion and exclusion criteria, as well as having complete data for all variables included in listed analyses. This study was approved by the Research Ethic Board at the Children's Hospital of Eastern Ontario.

2.2 | Participants

Children and adolescents with a concussion were only included if they were deemed eligible according to the following criteria: sustained a concussion within 48 h prior to the emergency department visit; were 10 through 17.99 years of age; and proficient in English. Participants with a concussion were excluded if they met any of the following criteria: Glasgow Coma Scale rating of ≤ 13 ; abnormality or incidental findings on standard imaging; required neurosurgical operative intervention, intubation, or intensive care; required sedation; incapable of resuming physical activity; history of trauma preceding injury.

Orthopedic-injured participants were included as a comparison group. Children and adolescents with orthopedic injuries are a suitable comparison group, as statistical comparisons with healthy non-injured children fails to eliminate confounds related to the experience of injury trauma. Although previous research identified similar alterations in DTI metrics in the acute phase of injury (i.e., <96 h post-injury; Wilde et al., 2019) between concussion and orthopedic-injured participants relative to healthy children, such similarities may be attributable to premorbid or predisposing risk factors (Ware, Yeates, Geeraert, et al., 2022; Wilde et al., 2019). Further, differences in DTI metrics between concussion and orthopedic-injured samples have been reported in the in the subacute and chronic phases (Roberts et al., 2014; Ware, Yeates, Tang, et al., 2022). Orthopedic-injured participants were included if they were 10 through 17.99 years of age; sustained an orthopedic injury isolated to the upper extremity within 48 h prior to the emergency department visit; and were proficient in English. Orthopedic-injured participants were excluded if they had a concussion within the past year or required sedation or surgical operation. As such, orthopedic-injured participants were eligible if they had a concussion over 1 year from the date of enrolment.

Any participants, concussion or orthopedic-injured, were excluded if they presented: communication difficulties from severe chronic neurological developmental delay; a history of neurological disorder; intellectual disability; intoxication during the emergency department visit or imaging session; documented sedation prior to the emergency department visit; history of hospitalization for psychiatric

illness; incapability of resuming physical activity; incapability of providing consent/assent; absent legal guardian; unavailable for data collection at both time points; medical contraindications to MRI; braces or metal implants. A total of 129 participants qualified for the MRI portion of the PEDCare study, 95 of which were included in this study after meeting set conditions.

Diagnostic history of depression and anxiety (i.e., history of psychiatric illness) and reported mechanism of injury was collected during the emergency department visit, while additional self-report data such as the Health Behavior Inventory (HBI; symptom scale) and Connor-Davidson Resilience Scale 10 were collected at both 72 h and 4-weeks post-injury. Each participant of this substudy underwent MRI scanning including a DTI sequence at both time points.

2.3 | Connor-Davidson resilience scale 10

The CD-RISC 10 (Campbell-Sills & Stein, 2007) is a 5-point Likert scale with 10 items, a shortened version of the original 25-item CD-RISC (Connor & Davidson, 2003). This scale computes a degree of resilience in the past month. This scale measures resilience by assessing characteristics such as the ability to adapt, maintain secure relationships, manifest confidence in face of new challenges, successfully cope with stress, and bounce back after hardship. As such, 72 h resilience may serve as a measure of pre-injury and acute resilience, while 4-week resilience may serve as a measure of acute and subacute resilience. The computed sum of all items on the CD-RISC 10 serves as a measure of psychological resilience, whereby higher scores indicate high resilience, and lower scores indicate low resilience. The CD-RISC 10 has good internal consistency, convergence, construct and content validity (Kuiper et al., 2019; Windle et al., 2011). Further, it has satisfactory construct validity with evidence of greater validity relative to the 25-item version when used in pediatric concussion populations (Durish et al., 2018). The scale has been previously used to assess longitudinal changes in resilience (Okuyama et al., 2018). For this study, the computed sum of all items on the scale, completed by the participant, was used.

2.4 | DTI acquisition

The DTI used a multi-slice 2D spin echo EPI sequence with the following parameters: resolution = $2.2 \times 2.2 \times 2.2 \text{ mm}^3$; slices = 57, FOV = $22 \times 22 \text{ cm}^2$, TR = 6300 ms, TE = 55 ms; $b = 0 \text{ s/mm}^2$, datasets, 30 directions at $b = 900 \text{ s/mm}^2$, 30 directions at $b = 2000 \text{ s/mm}^2$. Each DTI analysis incorporated volumes collected with $b = 0 \text{ s/mm}^2$, $b = 900 \text{ s/mm}^2$, and $b = 2000 \text{ s/mm}^2$ (Pines et al., 2020). The DTI scan duration was 7 min; identical sequences were used at both time points (72-h and 4-weeks post-injury).

The FMRIB software (FSL; Smith et al., 2004; Woolrich et al., 2009) was used to process the imaging data. A careful visual inspection of the images for signs of distortion determined that field distortion correction was not necessary for our sample. FA, AD, RD,

and MD were all assessed from the DTI data. FA was used to indirectly measure fiber density and organization. Although FA is highly sensitive to detecting structural changes in white matter, it lacks specificity as a biomarker of neuropathology and for the characterization of microstructure (Alexander et al., 2007). Thus, AD was used to assess axial microstructure, which is sensitive to axonal degeneration (Song et al., 2002). Meanwhile, RD was used to assess radial microstructure, which is sensitive to myelination (Song et al., 2002). MD, the mean of eigenvalues representative of AD and MD, was used to indirectly characterize changes in diffusion tensors (i.e., AD and RD). All DTI marker comparisons were computed using a voxel-wise statistical repeated measures analysis with a threshold-free cluster enhancement correction (TFCE) for regions of interest.

2.5 | Tract-based spatial statistics processing

Voxel-wise analyses of FA, MD, AD, and RD data were computed using tract-based spatial statistics following guidelines provided in the FMRIB software (Smith et al., 2006). All generated FA, MD, AD, and RD images were registered into a standard space using the mean FA/MD/AD/RD image and mean skeleton derived from the mean of all participants in each analysis. A mean skeleton image threshold of 0.2 was applied to each data set. All results were projected onto the MNI152_T1_1mm template. The Johns Hopkins University White-Matter Tractography Atlas was used to identify and label significant regions (Hua et al., 2008; Mori et al., 2005; Wakana et al., 2007). Specifically, the FSL software's automated atlasquery (*autoaq*) was used to identify the structures in the provided atlas to which the significant cluster peaks belong. If no structures were identified for a significant cluster, the cluster was not reported.

2.6 | Statistical analyses

Independent samples *t* tests were conducted to compare differences in age, resilience, and HBI scores between both groups. A chi-square test was used to determine whether the proportion of sex differed between groups. Paired samples *t* tests were conducted to differences in resilience and HBI scores between 72 h and 4-weeks post-injury. Outliers for resilience scores were defined as 2.48 standard deviations above or below the mean ($Z = + - 2.48$), based on recommendations following the effect of sample size on outlier detection (Cousineau & Chartier, 2010; van Selst & Jolicoeur, 1994). One outlier was identified within the concussion group for scores of 72 h resilience. One other outlier was identified within the concussion group for scores of 4-week resilience. The outliers were Winsorized to the nearest data point within their respective group (Dixon & Yuen, 1974). A total of three cases of 4-week resilience were missing within the concussion group. The 4-week resilience data were considered missing completely at random ($\chi^2 = 4.26[5, n = 66], p = .512$), and the expectation-maximization algorithm was applied to impute the missing 4-week resilience values (Dong & Peng, 2013). A total of 20 cases

of 4-week HBI data were missing within the concussion group. The 4-week HBI data were considered missing at random ($\chi^2 = 1.19$ [5, $n = 66$], $p = .946$) and was also imputed using expectation-maximization (Dong & Peng, 2013).

Nonparametric permutation testing was conducted on whole-brain FA, MD, AD, and RD values using FSL's randomize command with 5000 permutations in conjunction with a TFCE for each DTI metric (Winkler et al., 2014). The randomize command runs a permutation script that includes modeling and inferencing with the general-linear model design using the specified number of iterations per metric. The demeaning option was selected to demean the variables in the design matrices within each group. Clusters were extracted using a significance threshold of $p < .05$ TFCE corrected for multiple comparisons across space using nonparametric permutation testing (Smith & Nichols, 2009). Significant clusters were extracted with a cluster-extent threshold of ≥ 50 voxels.

The two-sample randomized pipeline was modeled to investigate the association between resilience and DTI metrics at both time points and whether group moderates the association. Further, this pipeline was used to examine whether 72 h resilience was associated to 4-week DTI metrics. These sets of analyses included the following covariates: (1) age, (2) sex, (3) history of concussion, and (4) history of psychiatric illness. Handedness was not included as a covariate due to insignificant differences in FA maps between healthy right- and left-handed children (López-Vicente et al., 2021). Covariates were entered jointly across the two groups to control for differences in white matter microstructure between groups. If significant interactions were detected in any of the statistical models, the relationships between resilience and the specified outcome (i.e., the regression slopes) were examined in each group separately on the corresponding white matter tract to further explain the interaction. The two-sample randomized pipeline was also used to investigate the association between resilience and DTI metrics within both groups separately. The analysis for the concussion group included the following covariates: (1) age, (2) sex, (3) history of concussion, and (4) history of psychiatric illness. Meanwhile, the analysis for orthopedic-injury group included only (1) age and (2) sex as covariates due to the smaller sample size.

3 | RESULTS

3.1 | Demographics

Of the 129 participants enrolled in the MRI component of the Ped-CARE study, 6 were excluded based on having left the emergency department without receiving instructions to join the study, 10 were excluded for missing behavioral data, 15 were excluded for missing imaging data, and 3 were excluded due to clinically significant incidental imaging findings (see Figure 1).

A total of 95 participants were included in this study, 66 of which were included as part of the concussion group (47% female, median age = 12.88 [IQR: 11.80–14.36]) while 29 were included as part of

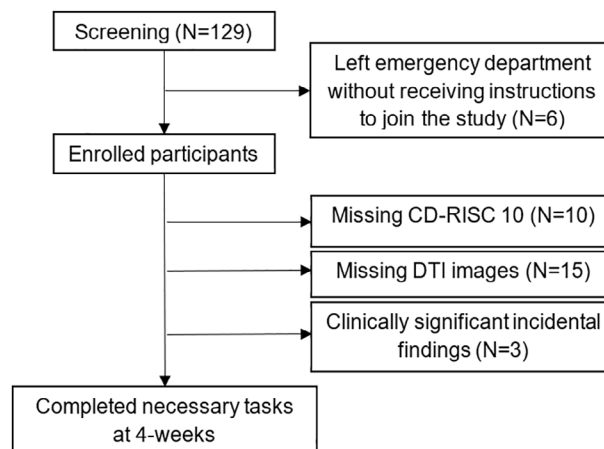


FIGURE 1 Participant flowchart

the orthopedic-injured comparison group (41% female, median age = 12.49 [IQR: 11.18–14.01]). The main mechanism of injury was sports for both the concussion group ($n = 37$) and the comparison group ($n = 19$). Both groups did not differ on age, history of concussion, sex, handedness, diagnostic history, 72 h post-injury resiliency scores, and 4-weeks post-injury resiliency scores (Table 1).

Independent samples *t* test found a significant difference in emergency department and 72 h HBI scores between groups, whereby the concussion group had higher scores than the comparison group ($t[93] = 3.85$, $p < .001$), ($t[93] = 5.34$, $p < .001$). No significant difference in 4-week HBI scores between groups was found ($t[93] = 0.13$, $p = .893$). HBI scores at 72 h post-injury were significantly greater than HBI scores at 4-weeks post-injury in the concussion group ($t[65] = 9.07$, $p < .001$). No significant differences were detected between HBI scores at 72 h post-injury and 4-weeks post-injury within the orthopedic-injured group ($t[28] = .45$, $p = .655$). Paired-samples *t* tests found no significant differences in 72 h post-injury and 4-week post-injury resiliency scores within the concussion group ($t[65] = -1.35$, $p = .181$) and the orthopedic-injured group ($t[28] = -0.155$, $p = .878$).

Permutation testing of between-group differences in DTI metrics found no significant cluster differences between the concussion group and orthopedic injury group at both 72 h and 4-weeks post-injury.

3.2 | Associations between resilience and DTI metrics

Analyses identified one significant inverse association between 72 h resilience and 72 h FA along the forceps minor (cluster size = 35,953; $MNI\ x\ y\ z = 17\ -16\ -11$; TFCE-corrected $p = .003$; $\beta = -2.60E-03$; 95% CI = $-5.37E-03$, $1.62E-04$; Figure 2). Two significant direct associations were identified between 72 h resilience and 72 h MD along the right superior longitudinal fasciculus (cluster size = 31,823; $MNI\ x\ y\ z = 20\ -2\ 12$; TFCE-corrected $p = .006$; $\beta = 1.80E-06$; 95%

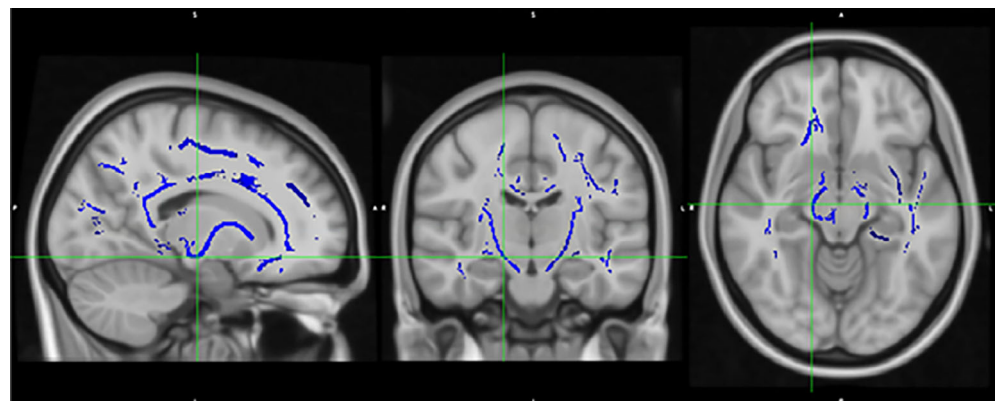
TABLE 1 Demographic data

	Concussion N (%)	Orthopedic injury N (%)	Group differences <i>p</i> -value
Age, median (IQR)	12.88 (11.80–14.36)	12.49 (11.18–14.01)	.265
History of concussion	16 (24.40)	6 (20.69)	.409
Sex			.614
Male	35 (53)	17 (59)	
Female	31 (47)	12 (41)	
Handedness			.102
Left	9 (14)	8 (28)	
Right	57 (86)	21 (72)	
Diagnostic history			
Anxiety	11 (17)	3 (10)	.423
Depression	4 (6)	1 (3)	.600
Attention-deficit/hyperactivity disorder	10 (15)	4 (14)	.863
Mechanism of injury			
Fall from elevation	1	1	
Fall down stairs	2	0	
Sport	37	19	
Fall from standing/walking/running	7	2	
Ran into stationary object	7	0	
Bike collision/fall while riding	0	1	
Other mechanism	12	6	

IQR: Interquartile range

	Concussion N (%)	Orthopedic injury N (%)	Group differences <i>p</i> -value
Resilience 72 h post-injury, median (IQR)	28 (20.75–31.25)	27 (21.50–30.50)	.621
Resilience 4-weeks post-injury, median (IQR)	28 (21.00–33.25)	28 (20.50–33.00)	.917
HBI at ED, median (IQR)	23 (16.00–30.25)	13 (7.50–23.00)	<.001
HBI at 72 h post-injury, median (IQR)	24 (17.00–32.00)	12 (3.00–20.50)	<.001
HBI at 4-weeks post-injury, median (IQR)	10.5 (1.75–17.50)	11 (3.00–18.50)	.893
Hours between ED and 72 h scan, median (IQR)	75.38 (53.26–93.69)	100 (68.77–119.96)	
Days between ED and 4-week scan, median (IQR)	30.24 (28.35–32.09)	30.70 (27.98–32.61)	

Abbreviations: ED, emergency department; HBI, Health Behavior Inventory; IQR, interquartile range.

FIGURE 2 Associations between 72 h resilience and 72 h fractional anisotropy (FA) in concussion group. Areas in blue reflect significant voxels at threshold-free cluster enhancement (TFCE)-corrected $p < .05$, whereby 72 h post-injury resilience is inversely associated with 72 h post-injury FA values. Cursor is on the maximum intensity voxel of the cluster, located on the right cerebral peduncle.

CI = $-1.23\text{E-}07$, $3.71\text{E-}06$; Figure 3) and right anterior thalamic radiation (cluster size = 426, MNI $x\ y\ z = 7\ -32\ -14$, TFCE-corrected $p = .048$, $\beta = 2.00\text{E-}06$, 95% CI = $-1.22\text{E-}07$, $4.13\text{E-}06$; Figure 3).

Analyses also detected one significant direct association between 72 h resilience and 72 h RD along the right superior longitudinal fasciculus (cluster size = 41,079; MNI $x\ y\ z = 37\ -36\ 8$; TFCE-corrected

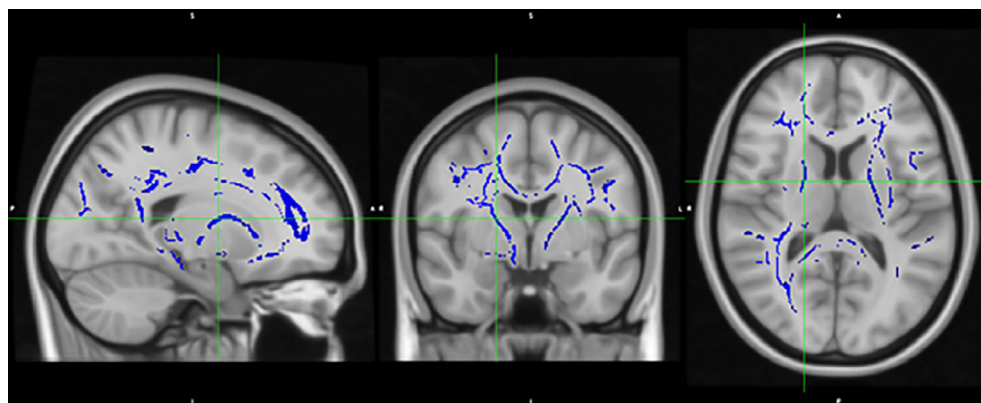


FIGURE 3 Associations between 72 h resilience and 72 h mean diffusivity (MD) in concussion group. Areas in blue reflect significant voxels at threshold-free cluster enhancement (TFCE)-corrected $p < .05$, whereby 72 h post-injury resilience is directly associated with 72 h post-injury MD values. Cursor is on the maximum intensity voxel of the cluster, located on the right posterior limb of the internal capsule.

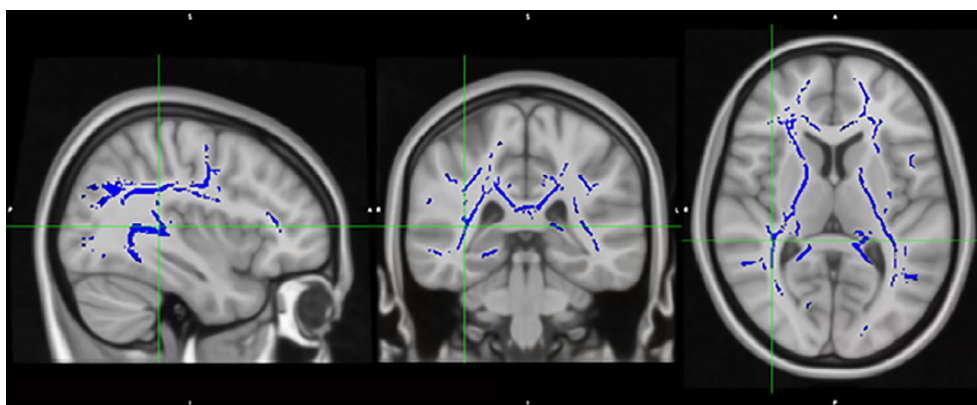


FIGURE 4 Associations between 72 h resilience and 72 h radial diffusivity (RD) in concussion group. Areas in blue reflect significant voxels at threshold-free cluster enhancement (TFCE)-corrected $p < .05$, whereby 72 h post-injury resilience is directly associated with 72 h post-injury RD values. Cursor is on the maximum intensity voxel of the cluster, located on the right retrolenticular part of the internal capsule.

$p = .004$; $\beta = 2.24E-06$; 95% CI = $-7.48E-08$, $4.55E-06$; Figure 4). No significant associations were identified between resilience and AD.

Groups did not moderate the association between 72 h resilience and 72 h DTI metrics, as well as 4-week resilience and 4-week DTI metrics. There were no significant clusters within orthopedic-injured children between 72 h resilience and each 72 h DTI metric following multiple corrections. There were no significant clusters within the concussion group between 72 h resilience and each 4-week DTI metric following multiple corrections. There were no significant clusters within the concussion group between 4-week resilience and each 4-week DTI metric following multiple corrections. There were no significant clusters within the orthopedic-injury group between 4-week resilience and each 4-week DTI metric following multiple corrections.

4 | DISCUSSION

In this study, we found significant associations between 72 h resilience and 72 h DTI metrics within children and adolescents with a concussion. We found no significant associations between 72 h resilience and 4-week DTI metrics, as well as between 4-week resilience and 4-week DTI metrics. There were no associations between 72 h

resilience and 72 h DTI metrics in the orthopedic-injured group, nor were there associations between 4-week resilience and 4-week DTI metrics. We did not find any significant interactions between group and resilience when predicting DTI metrics at both timepoints.

Our results suggest that injury type (i.e., concussion or orthopedic injuries) was not a significant moderator of associations between resilience and DTI metrics. Although resilience at each time point did not differ between groups, within-group comparisons of DTI metrics are not reported in this study. Schmidt et al. (2021) demonstrated that DTI metrics did not differ between young participants with concussion and orthopedic injury. Nonetheless, in support of our findings, the authors indicate that there were significant associations between resilience promoting factors and DTI metrics observed in the concussion group that were not identified in the orthopedic-injury group (Schmidt et al., 2021). Common pre-injury and injury-related factors between children with a concussion and orthopedic injuries may influence DTI metrics and ambiguous concussion-related alterations to white matter integrity. It follows that identified associations between resilience and DTI metrics might not be inherently unique to children with a concussion. Further research with a sample of healthy non-injured children is warranted to determine whether significant interactions exist between healthy controls and children with concussion or orthopedic injury when predicting outcome.

In partial support of our hypotheses, we found several associations between 72 h resilience and 72 h DTI metrics (FA, MD, and RD) within the sample of children with a concussion. Higher resilience was associated with higher MD and RD along the right superior longitudinal fasciculus. The superior longitudinal fasciculus connects regions of the temporoparietal junction and parietal cortex with the frontal cortex and is theorized to be implicated in language function (Dick & Tremblay, 2012; Wang et al., 2016). Higher resilience was associated with lower FA values along the forceps minor. The forceps minor connects regions of the frontal lobes by crossing the midline structures (Catani & Thiebaut de Schotten, 2008; Mamiya et al., 2018). Higher resilience was associated with higher MD along the right anterior thalamic radiation. The anterior thalamic radiation is a white matter bundle that connects regions of the frontal lobe to the thalamus and limbic system, thought to be implicated in executive functioning and negative emotion (Catani & Thiebaut de Schotten, 2008; Mamiya et al., 2018; Panksepp, 2005). These findings are consistent with a previous study reporting associations between resilience-promoting factors and both FA and MD in youth with a concussion, albeit in different white matter tracts (Schmidt et al., 2021). Schmidt et al. (2021) reported a positive correlation between resilience promoting factors and FA of the left sagittal stratum. The authors also report negative correlations between resilience promoting factors and MD of the anterior corona radiata and anterior limb of the internal capsule (Schmidt et al., 2021). There are few, if any, studies examining the associations between resilience and RD. We suggest that the reported associations between 72 h resilience and 72 h DTI metrics might be indicative of improved neural outcome in participants with higher resiliency. However, there is insufficient consistency in the literature pertaining to white matter injury progression in pediatric concussion to provide accurate inferences as to the integrity of white matter microstructure. Although FA, MD, AD, and RD are powerful indicators of white matter characteristics, including orientation and density, the variable effects of pediatric concussion on such metrics render it challenging to infer improved or worsened tract integrity (see Roberts et al., 2014 for a review). Here, we merely report that 72 h resilience is associated with the degree of directional diffusion (FA), magnitude of diffusion (MD), and degree of diffusivity across the axis of diffusion (RD) in white matter tracts in the acute phase of pediatric concussion.

Despite the significant clusters between 72 h resilience and 72 h DTI metrics in the concussion group, we did not find support for our hypothesis that 4-week resilience is associated with 4-week DTI metrics in either group. We also did not find any significant associations between 72 h resilience and 4-week DTI metrics in the concussion group. This might suggest that pre-injury/acute resilience is only an indicator of white matter microstructure in the acute phase of concussion. It is possible that concussion injury-related alterations to white matter microstructure in the subacute phase of concussion, including axonal swelling (Giza & Hovda, 2014; Roberts et al., 2014), ambiguate or eliminate the associations between resilience and 4-week DTI metrics. Longitudinal studies are thus required to examine whether time since injury is a significant moderator of resilience and DTI metrics in

pediatric concussion. Unfortunately, the nonparametric permutation methods applied in this study are designed for independent measures and are often not well-equipped for repeated measures (Winkler et al., 2014). The DTI data included in our study are inherently limited to nonparametric tests, as they do not meet the assumption of normality required for parametric tests. Hence, a longitudinal model was not feasible for our study but is advised for future investigations of resilience and white matter microstructure.

4.1 | Limitations

There is a paucity of consistency in DTI literature pertaining to directionality of FA, MD, AD, and RD outcomes following pediatric concussion. Directionality of DTI metric findings in pediatric concussion often varies when considering time since injury (Roberts et al., 2014). There is also considerable heterogeneity in methodology or timing of MRI acquisition in pediatric concussion research, further complicating predictions of findings (Dodd et al., 2014). Hence, caution is warranted when interpreting significant DTI findings regarding children with concussion. Future research should implement additional measures for the characterization of microstructural white matter integrity. Neurite orientation dispersion and density imaging, or NODDI, uses diffusion MRI to provide additional information regarding dendrites and axons such as the orientation dispersion index—a measure of angular variation across neurites (Zhang et al., 2012). This method of analysis was proven effective when investigating white matter microstructure following mild traumatic brain injury in adult samples (Churchill et al., 2017; Palacios et al., 2020). However, this method has yet to be used in a pediatric concussion sample. Future research is needed to address these limitations, offering evidence to demystify the complex relationship between resilience and white matter integrity following the shearing forces of concussion.

Healthy children and adolescents experience significant developmental changes in white matter integrity, including ongoing changes in myelination (Asato et al., 2010). Subtle differences in degree of myelination, axonal density, or fiber organization across developmental stages might be associated with changes in DTI metrics. In a study investigating the changes in white matter development from childhood to adulthood, Lebel et al. (2008) found that maximal levels of FA are established at different rates in different tracts. Most maximal levels of FA, for example, are established by 11 years of age in the corpus callosum and inferior longitudinal fasciculus (Lebel et al., 2008). While our groups were matched on age and the inter-quartile range for both groups was narrow, a large age range of 10–17.99 years may be considered as a potential limitation to our study. Structural differences, as measured with DTI metrics, can also be detected between male and female pediatric samples. White matter volume develops at a faster rate in boys than in girls, reflected by differences in FA (Johnson et al., 2014; Lawrence et al., 2023; Schmithorst et al., 2008; Wang et al., 2012). In short, children of different age and sex at time of scanning likely demonstrate differences

in FA, MD, AD, and RD maps. To control for such effects on our results, we implemented both age and sex as separate covariates in our statistical models. The comparison group of orthopedic-injured children was also matched to the concussion group in age and sex.

The self-reported resiliency data in our samples might be influenced by a diagnostic history of anxiety or depression. Some studies report that resilience was associated with decreased rates of depression in adults with adverse childhood experiences (Poole et al., 2017; Schulz et al., 2014). It follows that individuals with low levels of resilience have heightened risk of depression following adverse experiences. As such, cases of depression and anxiety in our pediatric sample might demonstrate lower self-reported resilience. Moreover, preexisting mental health conditions may influence the association between resilience and clinical outcomes following concussion (Gornall et al., 2021). As shown in a previous study, anxiety and depression were a significant moderator of the effects of resilience on post-concussion symptom severity in a sample of children with concussion (Durish et al., 2019). Our study did not discard cases with a diagnostic history of anxiety or depression, which may have influenced self-reported scores of resilience at each timepoint. However, our analyses of resilience as a correlate of DTI metrics across all timepoints within children with concussion included preexisting mental health conditions as a covariate, controlling for the effects of anxiety or depression as a confounding variable on the reported results. Further, it remains a possibility that pediatric injury type may impact self-reported resiliency. Truss et al. (2017) found that most children are resilient to post-traumatic stress symptoms following concussion, which might suggest that pediatric concussion might not have a significant impact on resiliency. In our study, resilience scores did not differ between orthopedic-injured children and children with concussion at both timepoints. This lack of differences in resilience scores within- and between-groups also restricts our interpretations of the results, inherently limiting the scope of this study. Further research with other samples (e.g., children with psychiatric illness) is required to determine whether there are meaningful differences in resilience scores that can be moderated by resilience-targeted clinical interventions.

5 | CONCLUSION

In this study, we identified significant associations between 72 h resilience and 72 h DTI metrics within children with a concussion. No interactions were identified between group and resilience when predicting DTI metrics at both timepoints. These findings warrant further longitudinal studies exploring the associations between resilience and white matter microstructure in the acute and subacute phase of pediatric concussion. Although DTI is a powerful tool, implementing additional imaging techniques, including functional connectivity and cerebral blood perfusion rates, may further clarify the interaction between injury type and resilience on white matter integrity. Future research should seek to further investigate associations between resilience and white matter microstructure, particularly whether resilience is associated to improved neural recovery following pediatric injury.

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CONFLICT OF INTEREST STATEMENT

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DATA AVAILABILITY STATEMENT

The data cannot be shared due to legal, ethical, or privacy restrictions.

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